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	Functional recovery following brain and spinal cord injuries and neurodegenerative diseases is likely to require the transplantation of exogenous neural cells and tissues, since the mammalian central nervous system (CNS) has little capacity for self-repair. However, neural cell or tissue transplantation is limited by the lack of tissue donors and the low survival rate of grafted cells. There is a critical need for an alternative approach to building biological substitutes, such as three-dimensional (3D) in vitro culture of neural cells to repair or replace the function of damaged nerve tissues. Our strategy is to place dissociated neural stem cells onto naturally-derived or synthetic hydrogel scaffolds and to culture the resulting cell-hydrogel constructs in a RWV bioreactor. The in vitro system we propose here includes three components: 1) primary neural stem cells isolated from embryonic rat cortical neuroepithelium; 2) naturally-derived and synthetic or biosynthetic hydrogels modified by brain ECM/adhesive proteins and loaded with growth factors, that promote cell survival, growth and differentiation; and 3) RWV bioreactors that offer an optimized microgravity environment and regulate tissue development. Our working hypothesis is that rotating bioreactors offer relatively well-controlled process		

Task Description:

parameters and reduced hydrodynamic forces that may regulate neural stem cell proliferation, neurogenesis and synaptic formation. Rotating bioreactors create low shear stress and randomized gravitational vectors that optimize cell recruitment. They also provide high supply rates of nutrients and gases that are expected to improve neural tissue reconstruction in our in vitro system. The proposed cell-hydrogel-bioreactor system may allow us to study and thereby control neural tissue growth and neuronal network formation under simulated microgravity conditions. A successfully engineered nerve tissue may have potential use in tissue replacement therapy for brain and spinal cord injuries and neurodegenerative diseases. Our studies of neurogenesis and neuronal network formation in the cell-hydrogel-bioreactor system may lead to a tissue model for potential space experiments in the future. The long-term objectives of this project are to understand how microgravity and reduced hydrodynamic stress affect neural tissue generation from neural stem cells, and to develop a general approach to engineering functional neuronal networks for potential use in neural tissue replacement therapy. To accomplish these goals, four specific aims are proposed. Aim 1 is to develop or optimize methods to use naturally derived collagen and synthetic hydrogels such as biodegradable polymer, Poly(lactic acid) modified by extracellular matrix proteins for scaffold preparation. In Aim 2, we seek to immobilize neural stem cells and progenitors in the hydrogel scaffolds. We will characterize the resulting cell-hydrogel interface by examining cell adhesion and integrin-mediated signal transduction. Aim 3 is to characterize neurogenesis in simulated microgravity. We will examine whether hydrodynamic conditions in RWV bioreactors affect neural stem cell proliferation. We will then assess the effects of reduced hydrodynamic stress and microgravity on neural cell lineage progression and neuronal production. In Aim 4, we will characterize synaptogenesis and functional neuronal network formation in simulated microgravity. We reported the first engineered construct resembling embryonic brain tissue generated by the stem cell-collagen-bioreactor culture system. This novel strategy constructs multi-layered neural tissue whose stratification resembles that of the embryonic rat cerebral cortex. There are at least three major advantages to this strategy for neural tissue engineering. First, the use of neural stem and progenitor cells, which are capable of both self-renewal and phenotypic differentiation into all types of neural cells, has the potential for supplying different cell types under controlled conditions. Second, the use of collagen, which is the most widespread ECM protein in mammalian tissues, provides a scaffold for cell growth in three-dimensions. The matrix is useful for the growth of anchorage-dependent cells like neural stem and progenitor cells into neuronal circuits and networks. Third, RWV bioreactors provide simulated microgravity that creates low fluid shear stress and randomized gravitational vectors. Compared to cultures under static 1g condition, the RWV-generated tissues contain cells with significantly higher rates of proliferation and survival. The majority of cells aggregating into spheres in static cultures died in 4-5 weeks while RWV cultures promoted compact aggregates, which became transformed into a multi-layered tissue without a necrotic core. Thus far, the functional engineered tissue-like constructs can be maintained in vitro for up to 9 weeks. This method demonstrates the promise of utility of tissue engineered constructs not only for in vitro studies of neural stem cell biology, but also for a tissue replacement strategy with new functional brain-like tissues.
The novel approach described here would be amenable to build different neural tissue models depending on controlled conditions. For example, a combination of bFGF, EGF and LIF would be a potent synergistic stimulation of neural stem cell proliferation, that generates a continuously proliferating population of immature cells. In order to generate more excitable neurons and active synaptic activity, an early withdrawal of bFGF and addition of BDNF are necessary. Furthermore, the method described herein would be amenable to build other types of tissue models. As shown previously, a cell-polymer-bioreactor system was used to promote chondrogenesis and cardiac tissue formation. This strategy could also be used to generate functional CNS tissue constructs derived from neural stem and progenitor cells from a variety of species including humans. Thus, in the future the in vitro model could be scaled-up for clinical use. In addition, this method can be applied to the generation of a cheap and reproducible in vitro system with greater control of variables for the testing of drugs and for detection of environmental toxins.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Work in the FY2005 focused on two aspects. First, we explored the molecular mechanisms underlying the formation of two-layered, embryonic brain-like constructs generated in our cell-collagen-bioreactor system. Secondly, we used real time RT-PCR to analyze the expression of genes encoding TuJ1, GFAP and nestin during 3D cultures under RWV and static conditions.
Basic FGF/FGFR signaling is required in two-layered structure formation

We described in the FY04 report that the characteristic two-layered construct resembling embryonic brain tissue can be generated using the neural stem cell-collagen-bioreactor system. In the engineered construct, neural stem and progenitor cells actively proliferate and differentiate into neurons, astrocytes and oligodendrocytes, and form a complex three-layered structure that superficially emulates the cerebral cortex of the embryonic brain. The surface layer composed primarily of neural progenitor cells which predominantly express functional neurotransmitter receptors for cholinergic and purinergic agonists while differentiating cells in the deeper layer mainly respond to GABAergic and glutamatergic agonists and to veratridine. The mechanism by which the collagen-entrapped cells are polarized during the rotary culture is unknown. However, increasing evidence in developmental biology has shown the importance of biochemical environment through the discovery of morphogen gradients. Morphogens are substances that are secreted by specialized cells in the embryo. They degrade as they diffuse out from their source cell, and subsequently serve as a landmark for other cells. By sensing the concentration of a morphogen, a cell can determine its location, and this can in turn affect the fate of the cell. We hypothesize that bFGF and FGF receptors may play a key role in cell polarization during 3D culture in rotary system. Immunostaining with antibodies against bFGF, FGFR-1 and FGFR-3 was carried out in sections cross cell-collagen constructs. We found significant bFGF and FGFR1 and 3 expressions in an outside-in gradient across the construct. Basic FGF and FGFR-1 staining was most prominent within the layer I, and FGFR-3 staining was also detectable in layer I. The immunostaining was downregulated toward layer II. It was even more interesting that FGFR antagonist PD173074 prevented layer formation. PD173074 was added to the culture medium at concentrations of 100 nM and 1 ?M. Immunostaining for TuJ1, PCNA and nestin show that the layer I formation is partially interrupted by addition of 100 nM PD173074, and completely blocked by 1 ?M PD173074. These results indicate that bFGF/FGFR signaling is required for layered formation.

RWV promotes TuJ1, GFAP and nestin gene expression in 3D cultures

Changes in relative quantities of TuJ1, GFAP, and Nestin gene expression were measured during 3D cell cultures using real time RT-PCR. Total RNAs were isolated from 3D cell cultures. The fold change in TuJ1, GFAP, and Nestin relative to the -bactin endogenous control was determined by fold change=2(-D(DCT), where DCT=CT, target-CT,

	b-actin and D(DCT)= DCT, 3D cultures-DCT, D8-static(D21 static for GFAP). CT represents threshold cycle number. The gene expressions among three dimensional cultures at 1, 3 and 6 weeks were compared between static and RWV culture conditions. Real time RT-PCR analysis showed that expression of genes encoding TuJ1, GFAP and nestin was significantly lower in RWV than in static cultures in the first week. However, Gene expression was significantly higher in RWV than in static cultures at 3 and 6 weeks. These results are consistent with those from immunocytochemistry. These results indicate that RWV promoted TuJ1, GFAP and nestin gene expression in a long-term culture while a short RWV exposure (about one week) resulted in the retardation of the gene expression. Significance A fundamental challenge for tissue engineering is to understand organogenesis, the process by which cells self-assemble into tissues and then organs. Researchers are only beginning to compile the molecules that control cell-cell interactions and cell matrix remodeling, and that stimulate migration, proliferation, differentiation and apoptosis of cells in appropriate locations and sequence to form functional multicellular organs. Using the neural stem cell-collagen-bioreactor system, we are able to form an embryonic brain-like construct through neural stem cell development. During this engineering process, bFGF/FGFR signaling is required. Our results have also shown that simulated microgravity (RWV) induced significant disturbance in neural stem cell gene expression, growth and differentiation in the first week. This would affect CNS development and regeneration not only in embryos within pregnant women and children, but also in adults, since there are a significant number of neural stem and progenitor cells in adults. Therefore, our studies indicate potential central nervous system risks caused by space flight. Our in vitro 3D model provides not only a research tool but also the utility in tissue replacement therapy for the injured brain and spinal cord and in neurodegenerative diseases. In addition, the engineered functional tissue construct may serve as tissue surrogate for drug screening and detection of environmental toxins.
Bibliography Type:	Description: (Last Updated: 12/04/2005)
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